

BCE-C611
SEMESTER EXAMINATION-MAY 2023
CLASS:B.TECH SEMESTER: VI
DISTRIBUTED SYSTEMS

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A
(SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
(5 X 6 = 30 Marks)

- Question-1: What is a distributed file system and how does it differ from a traditional file system?
- Question-2: What is Domain Name and how it works in the distributed system. List the limitations of the Distributed systems.
- Question-3: What are some challenges that arise when designing distributed systems?
- Question-4: What is marshallng and why is it important in interprocess communication?
- Question-5: What is the role of the operating system layer in supporting distributed objects and remote invocation?
- Question-6: What is the role of name services in a distributed file system and how do they help to locate and access files?
- Question-7: What are some challenges involved in achieving distributed mutual exclusion?
- Question-8: What is the difference between physical time and logical time in a distributed system?
- Question-9: What is the difference between sequential consistency and release consistency in distributed shared memory systems?
- Question-10: How client server communication is possible in distributed systems?

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 X 10 = 40 Marks)

- Question-1:** Explain the architectural models of distributed systems, highlighting the advantages and disadvantages of each.
- Question-2:** Discuss design issues of DSM in detail.
- Question-3:** Discuss the authentication framework in Distributed System.
- Question-4:** Using the case study of Interprocess Communication in UNIX, describe the different types of interprocess communication mechanisms available in UNIX and their relative strengths and weaknesses.
- Question-5:** Describe the various mechanisms that operating systems use to support distributed objects and remote invocation. In your answer, discuss the role of the OS layer, protection, processes, threads, and communication, and how these mechanisms contribute to the overall OS architecture.
- Question-6:** Explain Distributed Mutual Exclusion and its types in detail.
- Question-7:** Discuss the concepts of clocks, events, and process states in a distributed system. Explain the role of physical clocks and the challenges involved in synchronizing them and describe how logical clocks can be used to establish a partial ordering of events. Using the topic of distributed debugging, explain how global states can be used to analyze the behavior of a distributed system.
- Question-8:** Explain the Common Object Request Broker Architecture (CORBA) and its role in distributed systems. Discuss the CORBA Remote Method Invocation (RMI) framework, including its design and implementation, and the services that it provides. Using a case study, discuss how CORBA is used in practice to build distributed systems.